



InfiniBench: Infinite Benchmarking for Visual Spatial Reasoning with Customizable Scene Complexity

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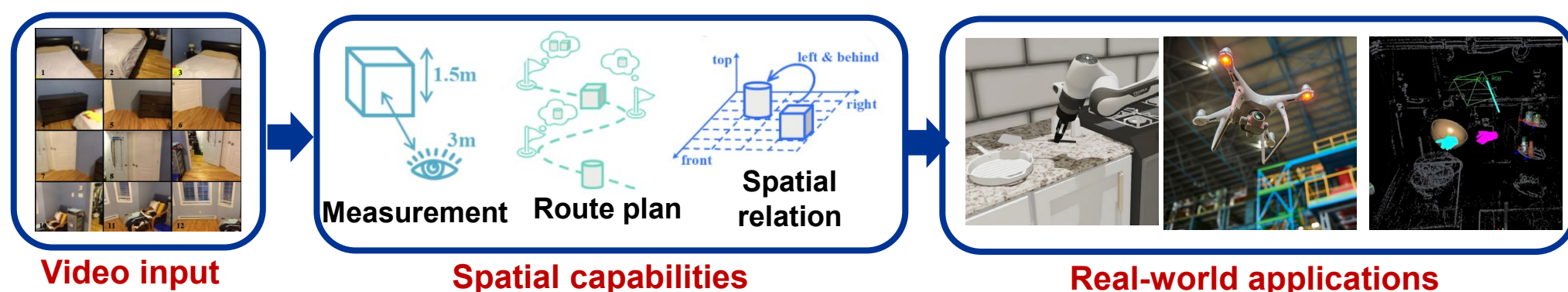


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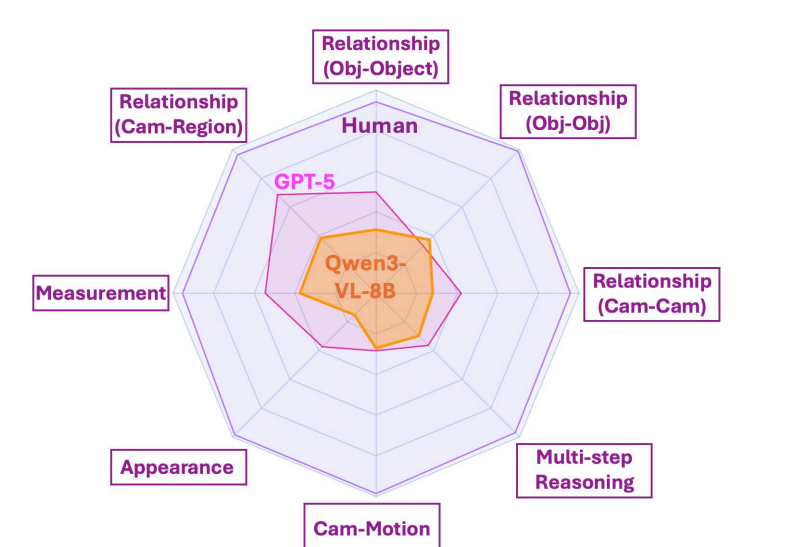
Background and Motivation

❖ Background: Spatial Reasoning for Vision Language Models (VLMs)



❖ Performance of current VLMs:

❑ Far Behind humans



❑ Struggle in complex scenes



Simple scenes

- Need to benchmark VLMs under different levels of scene complexity

Complex scenes

❖ Customizable Scene complexity in multiple dimensions

Compositional Relational Observational

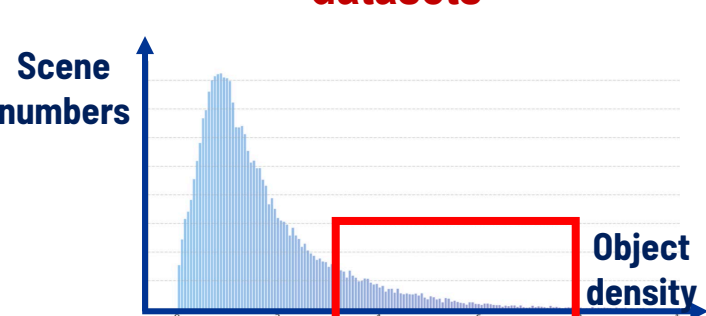


Existing benchmarks: coarse grained definition of scene complexity (e.g. defined by the number of room)

Our goal: Each dimensions of the complexity can be controlled independently

❖ Existing methods to achieved customizable scene complexity

❑ Search in existing datasets



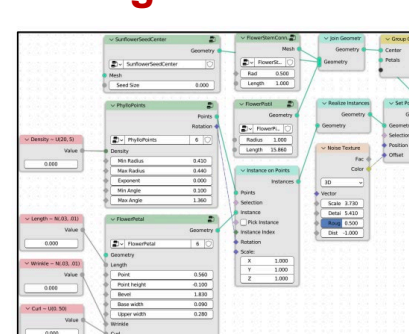
- Lack of high complexity scenes

❑ LLM based layout generation



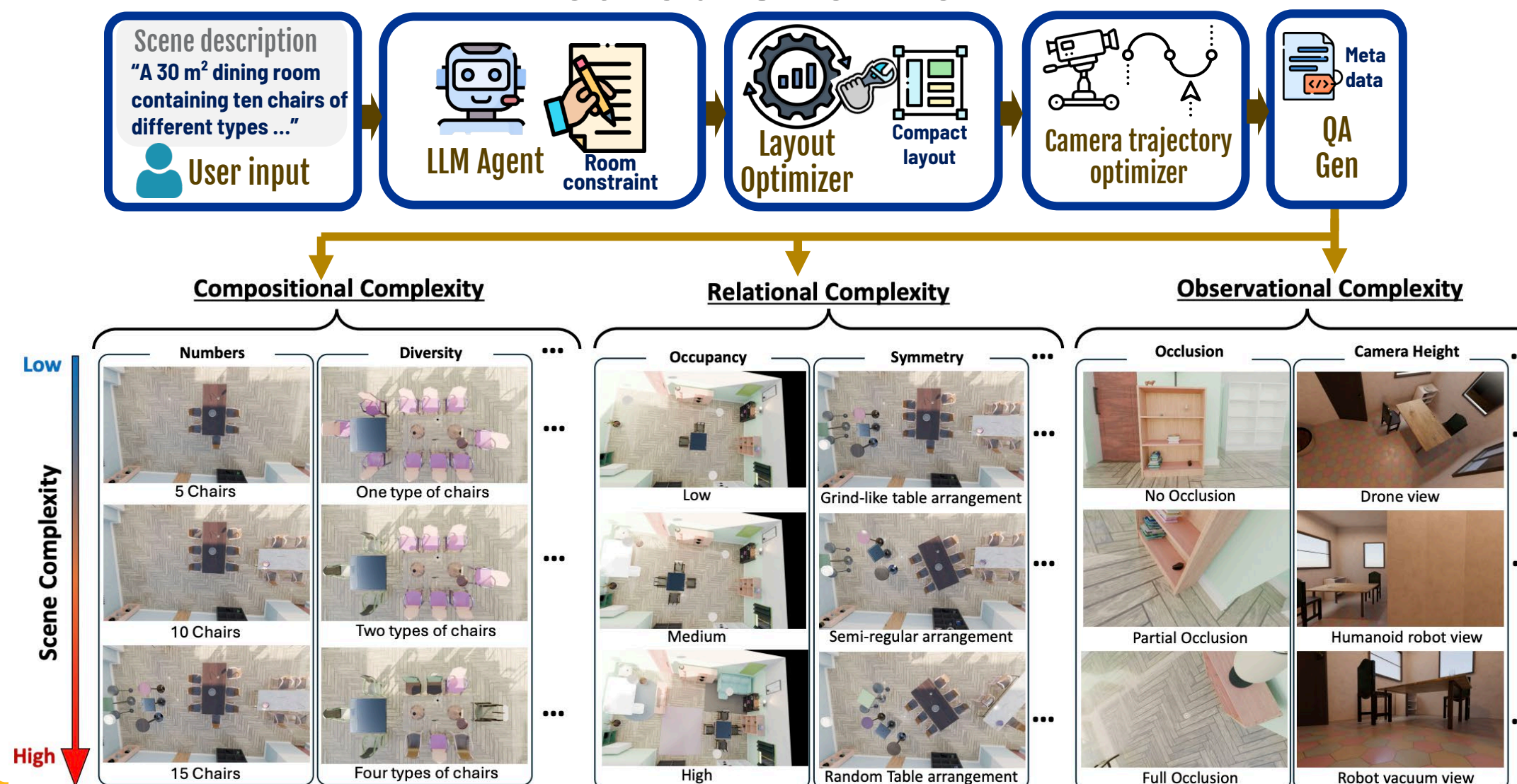
- Generated layouts may be physically invalid

❑ Procedure scene generation



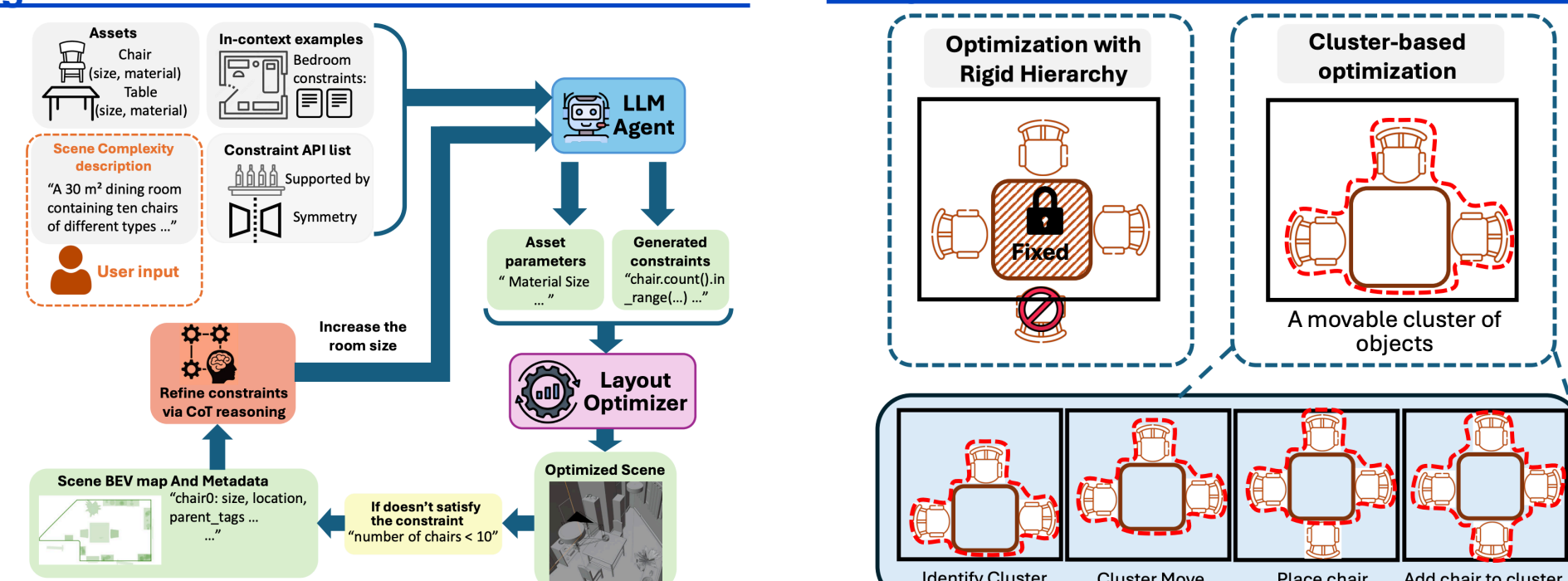
-Difficult to use
-Fail to generate complex scenes

Method Overview

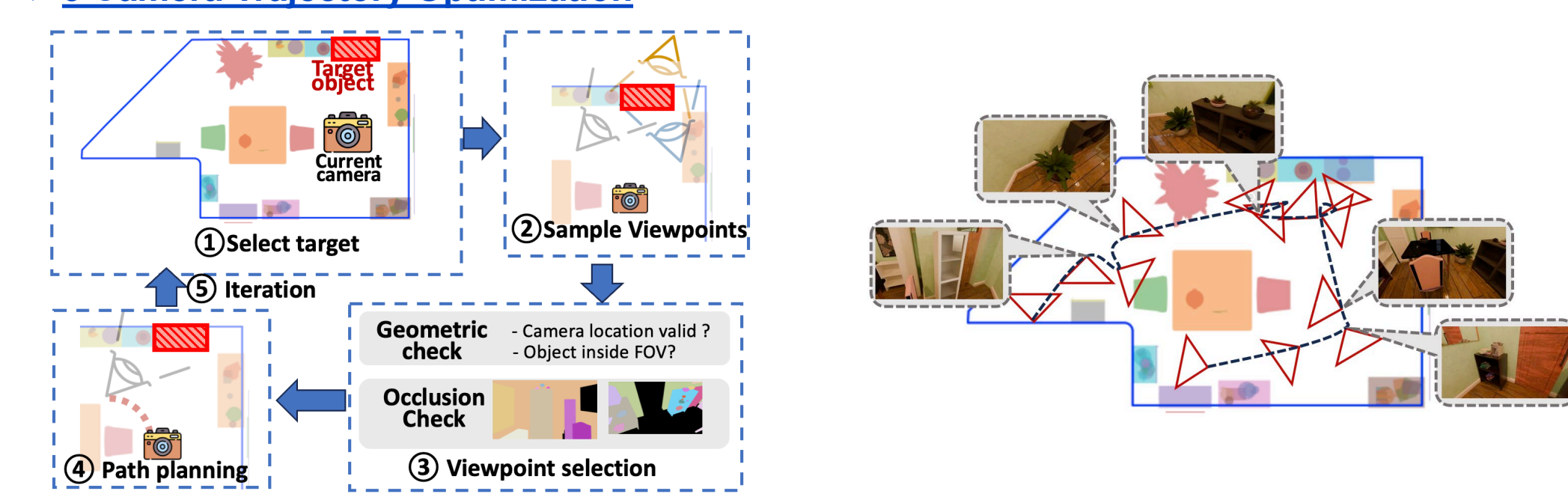


Method Details

❖ 1 Agentic Generation of Scene Constraints ❖ 2 Layout Optimization for Complex Scenes



❖ 3 Camera Trajectory Optimization



Evaluation of Scene generation

Dining room with 18 chairs around tables, and several shelves and appliances along the wall.

Classroom with 15 chairs and 2 sofas. The chairs should face the monitor on the desk, while the sofas should be placed against the wall with a small table nearby.

Bedroom with a bed facing a TV on a TV shelf, a chair facing a monitor on the desk, and several shelves along the wall.

LLM based generation **Procedural generation** **Ours**

• **LLM-based methods** achieve target scene complexity and high fidelity but suffer from more collisions and physical implausibility as complexity increases.

• **Procedural methods** maintain physical plausibility but struggle to satisfy complexity requirements, leading to lower fidelity in complex scenes

Using InfiniBench: diagnosing VLM failures

❖ **Object counting under different compositional complexity**

❖ **Observational complexity (camera height)**

BEV view

Ego-centric

Less objects More objects

Performance

Object Number

VLM reasoning performance declines as the number of objects increases, likely stems from repetitive counting across frames, which often leads to object overestimation.

Perspective-taking and spatiotemporal performance improves with higher viewpoints, likely because bird's-eye views provide a clearer and more complete representation of the scene's spatial layout.